

name: _____

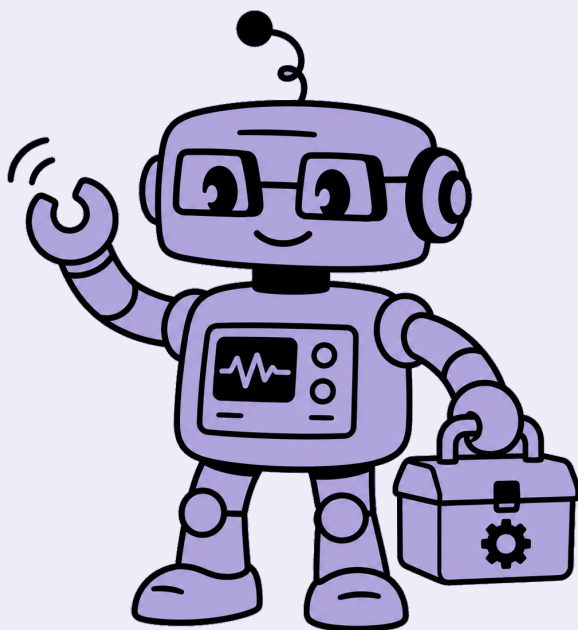
openhouse

robotics • level 2

✦ electronics ✦

your experience & progress book

ages 5-8



the four skills

the skills you grow

in every machine you grow the same four skills. look for the little icon at the top of each page – it tells you which skill you're using.



building & making

you build a real machine from blocks and circuit cards, step by step.



observing & understanding

you watch what your circuit does – then work out *why* it works.



problem solving

you think up and design your own ways to make it better.



presenting

you show your machine and explain how it works.

table of contents

	machine	the big idea	page
1	railway barrier	open & closed circuits	p.5
2	wind turbine	turning motion into electricity	p.13
3	soccer bot	reversing a motor	p.21
4	cleaning bot	sharing power between jobs	p.29
5	sensor-controlled crane	input and output	p.37

each machine takes you through all four skills – build it, watch it, understand it, improve it, and present it.

before you begin

staying safe with electricity

electricity is a helper, but we treat it with care. four rules for every engineer:



the battery blocks in your kit are **safe** to touch and build with – never use wall plugs or house electricity.



never put wires or blocks in your mouth, or near water.



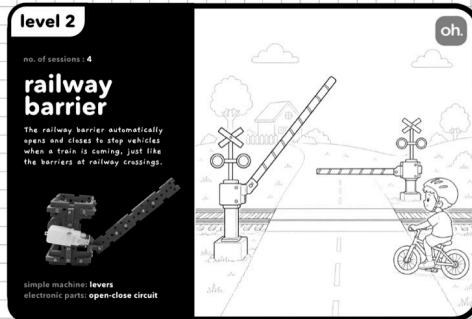
if a wire feels warm, switch it off and tell your teacher.



always build with your teacher nearby – engineers work safely.



meet the railway barrier



your railway barrier – it lifts and lowers when you flip the switch.

Every morning, Aarav's school bus stops at the railway crossing. A striped barrier comes down, a bell rings, and everyone waits as the train rushes past. Then the barrier lifts and the bus rolls on. how does it know when to go up and down? today, think like an engineer.

words to know

circuit

switch

current



did you know?

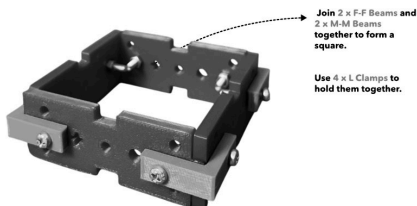
the flow of electricity going round the loop is called **current**.



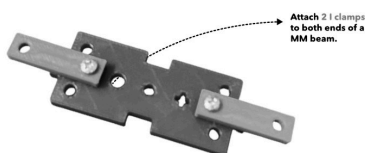
building & making

build your railway barrier – follow the steps.

step 1 : assemble the chassis

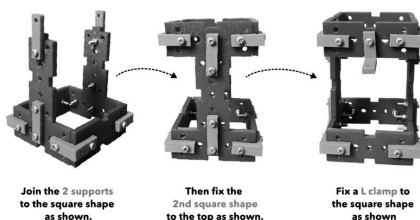


step 2: assemble the supports



Repeat the step to make another one.

step 3: join the supports to the chassis



tick when you've built the frame

next: add the barrier arm and the switch, so it lifts and lowers when you flip the switch.



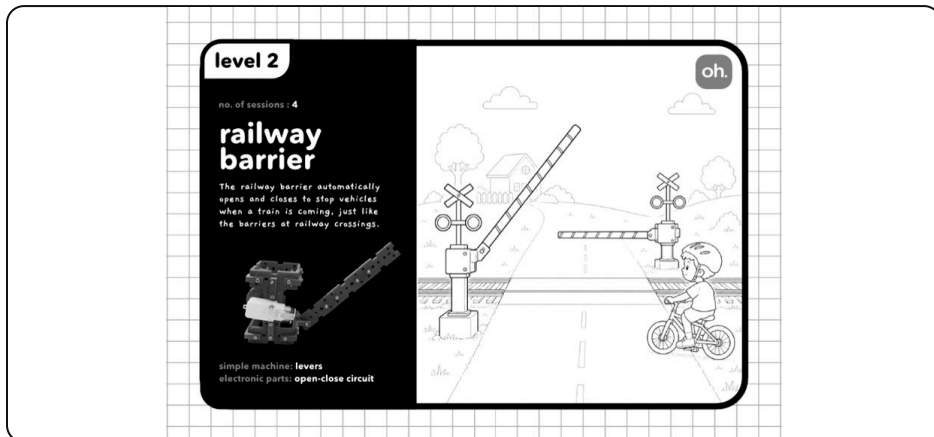
building & making



name the parts



on the picture, circle each part and write its name next to it.



label these four parts

the motor

the battery

the switch

the barrier arm

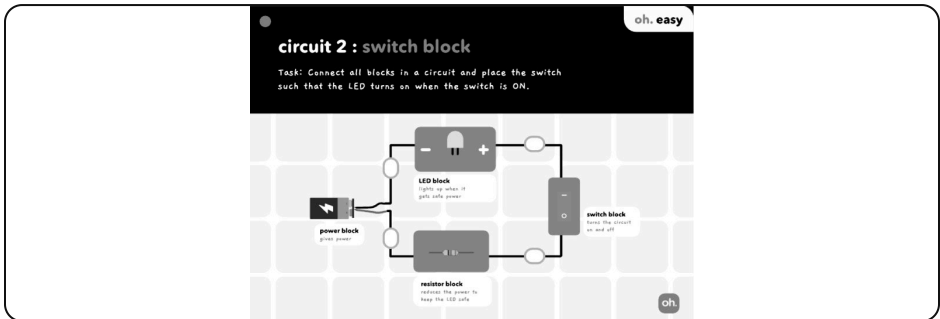


bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when you pressed the switch?

- ☐ the light came on when the switch was on, and went off when it was off
- ☐ the light kept blinking by itself
- ☐ the light stayed off the whole time

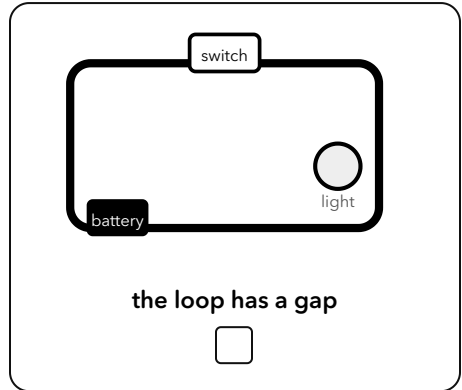
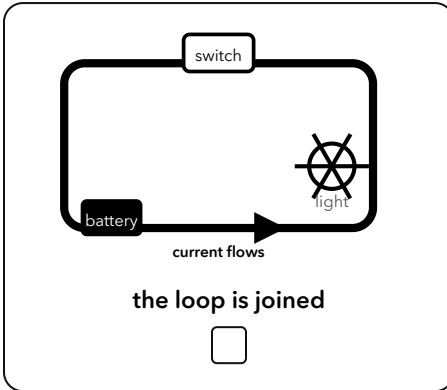


draw what you saw



observing & understanding

the big idea – why does it work?



the switch opens and closes the loop. when the loop is **joined**, the **current** flows all the way round (see the arrows) and it works. when there's a **gap**, nothing flows.

🔧 can you finish it?

for the barrier to move, the loop must be...

- ☐ joined all the way round
- ☐ broken with a gap in it



problem solving



how could you make it even better?

 circle the idea that works best.

1 • make the crossing easy to see at night

so drivers slow down in the dark. which idea works best?

- ☐ add a bright light to the barrier
- ☐ paint it a dull grey
- ☐ make it thinner

2 • warn people before the barrier closes

how could it tell them to step back?

- ☐ make it beep and flash first
- ☐ close as fast as it can
- ☐ give no warning

3 • keep the control box working in the rain

what would protect it best?

- ☐ put a little roof over the box
- ☐ pour water on the wires
- ☐ leave it out in the rain



your idea: think of one more way to make it better. draw it and label it.

 draw your idea here



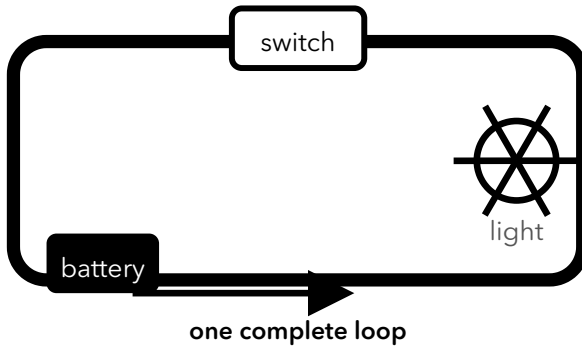
problem solving



engineer challenge – design your own

you want a light that switches on when the barrier closes, to warn cars.

here's an example – a switch turns a light on and off:



🔧 tick the blocks you would need

power block

switch block

light block

motor block

wires



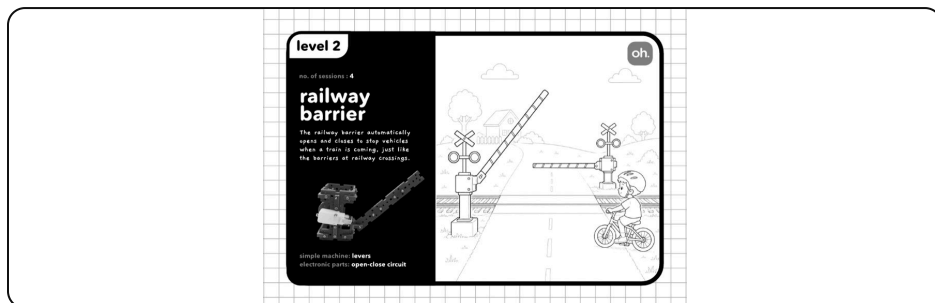
now draw your own circuit – join the blocks in a **complete loop** so it works.

🔧 draw your circuit here



presenting

can you be the teacher?



show a partner your railway barrier. explain:

1. what does it do?
2. which part gives it power?
3. which part makes it work?

★ **challenge:** show them how to raise and lower it.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it keeps cars and people safe at the railway crossing.)

family member: _____

signature: _____



meet the wind turbine



your wind turbine – it spins in the wind and lights up an led.

On a windy hill, Meera watches a giant white windmill turn, slow and steady. Her teacher says it isn't grinding grain – it's making electricity for the whole village. Meera wonders: how can spinning blades light up a bulb? today, you'll find out.

words to know

energy

led

generator



did you know?

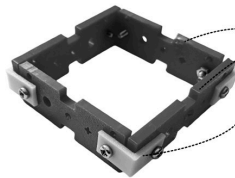
a spinning motor can **make** electricity – then it's called a **generator**.



building & making

build your wind turbine – follow the steps.

step 1 : assemble the base

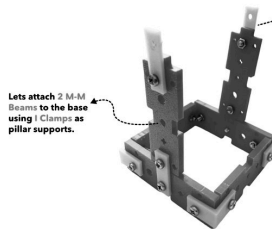


Let us start with building the base of the Wind turbine.

Join
2 x F-F Beam and
2 x M-M Beams
together to form a square.

Use 4 x L-Clamps to hold
them together and
tighten it with screws
and nuts.

step 2 : assemble the pillar supports



Lets attach 2 M-M
Beams to the base
using 1 Clamps as
pillar supports.

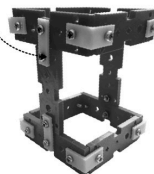
Fix the other set of
i-clamps to the top
of the pillars.

step 3: assemble the 2nd base support

repeat step 1 and
make the second base



now, fix the 2nd
base structure to
the top of the pillar
onto the i-clamps



tick when you've built the frame

🔧 next: connect the motor to the led block, so it spins in the wind and lights up an led.



building & making



name the parts

👉 on the picture, circle each part and write its name next to it.



label these four parts

the blades

the motor

the led

the wires

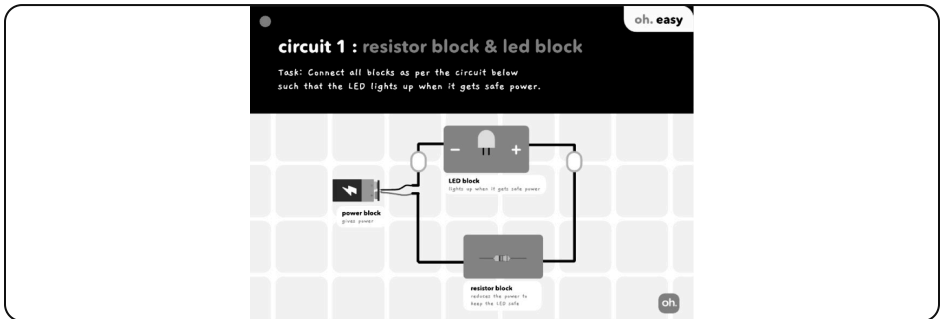


bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when the blades spun faster?

- ☐ the led glowed brighter
- ☐ the led went dark
- ☐ nothing changed at all

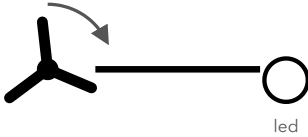


draw what you saw

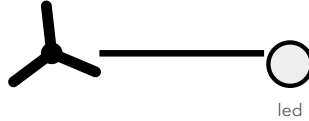


observing & understanding

the big idea – why does it work?



spinning fast



barely spinning

the spinning blades turn the motor into a tiny **generator** – it makes electricity. the **faster** it spins, the more power, and the brighter the led. (an led only lights one way round.)

can you finish it?

to make the led glow brighter, the blades should spin...

☐ faster

☐ slower



problem solving



how could you make it even better?

 circle the idea that works best.

1 • catch more wind so it spins faster

which idea would help most?

- ☐ make the blades bigger and turn them to face the wind
 - ☐ make the blades tiny
 - ☐ point them away from the wind
-

2 • light the led even in a gentle breeze

what could you try?

- ☐ spin it by hand to give it a start
 - ☐ wait for a storm
 - ☐ take the led out
-

3 • show how much power it makes

what would show it best?

- ☐ a bright led that glows more when it spins faster
 - ☐ a sticker
 - ☐ nothing
-



your idea: think of one more way to make it better. draw it and label it.

 draw your idea here



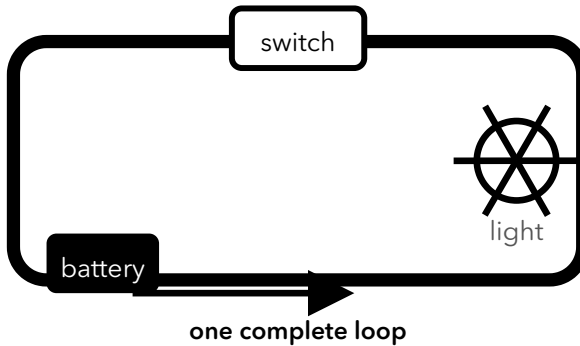
problem solving



engineer challenge – design your own

you want the led to light up when the blades spin.

here's an example – a switch turns a light on and off:



🔧 tick the blocks you would need

motor block

led block

resistor block

wires

switch block



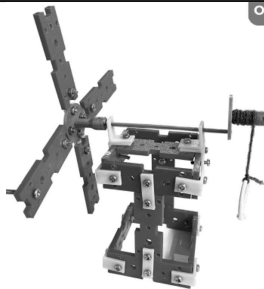
now draw your own circuit – join the blocks in a **complete loop** so it works.

🔧 draw your circuit here



presenting

can you be the teacher?



show a partner your wind turbine. explain:

1. what does it do?
2. which part gives it power?
3. which part makes it work?



challenge: show them how to spin it to light the led.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it turns wind into electricity to light a lamp.)

family member: _____

signature: _____



meet the soccer bot

oh.



your soccer bot – it drives forward and back when you flip the switch.

It's the big match! Kabir grips his controller as his little robot races across the floor, nudges the ball, and – GOAL! But to defend, he has to make it reverse, fast. how can one robot go both forwards and backwards? that's your challenge.

words to know

motor

direction

reverse



did you know?

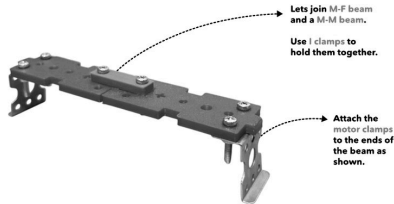
swap the two wires on a motor and it spins the **other** way – that's **reversing**.



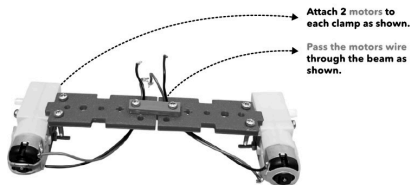
building & making

build your soccer bot – follow the steps.

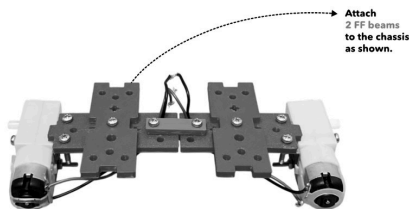
step 1 : assemble the chassis



step 2: attach the motors



step 3: complete the chassis assembly



tick when you've built the frame

🔧 next: wire the motors through the dpdt switch, so it drives forward and back when you flip the switch.

soccer bot



for the child



building & making



name the parts

👉 on the picture, circle each part and write its name next to it.

oh.



label these four parts

the motor

the wheels

the battery

the switch

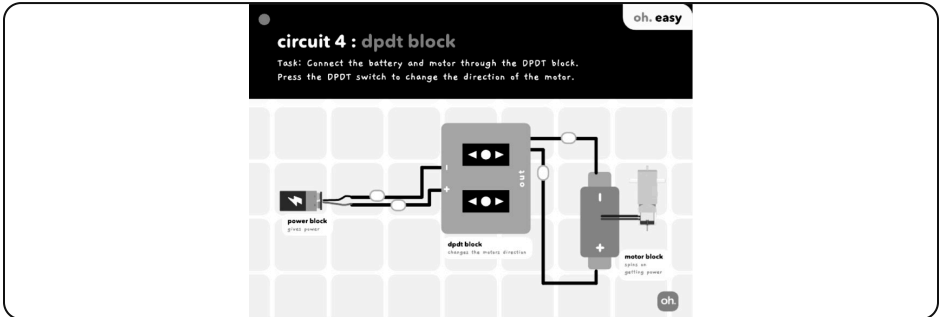


bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when you flipped the dpdt switch?

- ☐ the motor spun the other way, so the bot reversed
- ☐ the motor stopped forever
- ☐ the wheels fell off

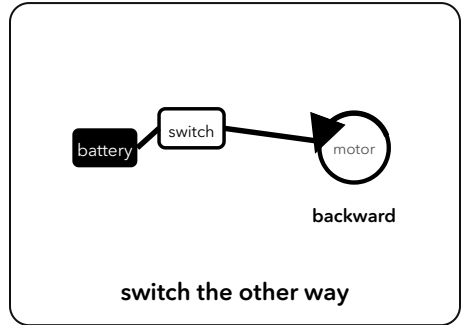
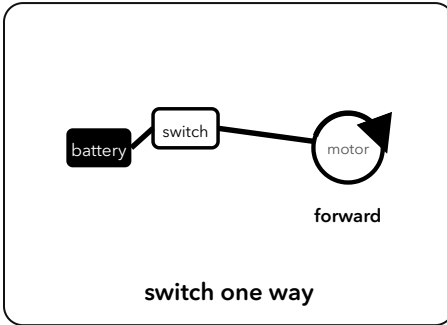


draw what you saw



observing & understanding

the big idea – why does it work?



flip the switch and the electricity flows the **other way** – so the motor spins backwards. that's how the same bot goes **forward and back**.



can you finish it?

to make the bot reverse, you...

- ☐ flip the switch the other way
- ☐ add more wheels



problem solving



how could you make it even better?

circle the idea that works best.

1 • make the bot turn to chase the ball

which idea would work?

- ☐ give each wheel its own motor and switch
- ☐ glue the wheels still
- ☐ remove a wheel

2 • make it quicker off the mark

what could help?

- ☐ a fresh, fully-charged battery
- ☐ a flat battery
- ☐ no battery

3 • stop it running off the pitch

what would help the player?

- ☐ a switch to turn it off fast
- ☐ no switch at all
- ☐ tape over the wheels



your idea: think of one more way to make it better. draw it and label it.

draw your idea here



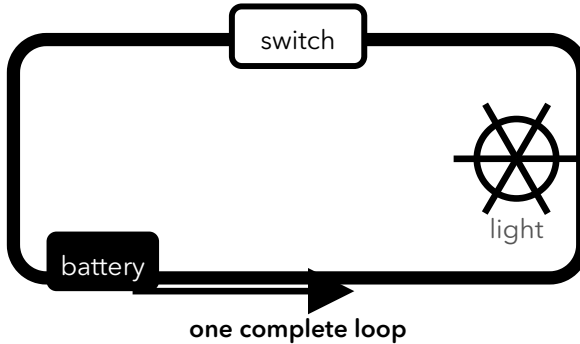
problem solving



engineer challenge – design your own

you want a switch that makes the bot go forward and back.

here's an example – a switch turns a light on and off:



🔧 tick the blocks you would need

power block

dpdt switch

motor block

wheels

wires



now draw your own circuit – join the blocks in a **complete loop** so it works.

🔧 draw your circuit here



presenting

can you be the teacher?

oh.



show a partner your soccer bot. explain:

1. what does it do?
2. which part gives it power?
3. which part makes it work?



challenge: show them how to drive it forward and back.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (you drive it to chase the ball and score goals.)

family member: _____

signature: _____



meet the cleaning bot



your cleaning bot – it drives along and sweeps at the same time.

Diya's room is a mess again! She wishes for a little robot that could drive around AND sweep at the same time. Her uncle smiles: one battery can do more than one job at once. how? today, you'll build it and see.

words to know

power

parallel

current



did you know?

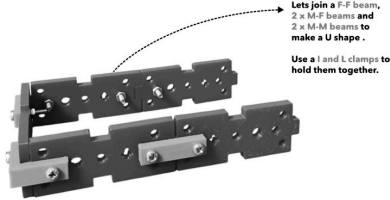
when one battery runs two things at once, we say they are wired in **parallel**.



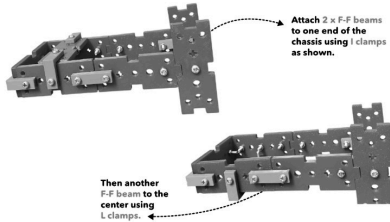
building & making

build your cleaning bot – follow the steps.

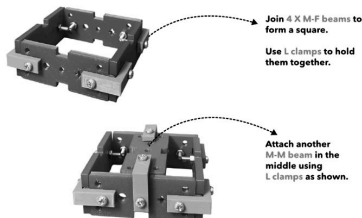
step 1 : assemble the chassis



step 2: attach the supports



step 3: assemble the front chassis



tick when you've built the frame

🔧 next: wire both motors to the one battery, so it drives along and sweeps at the same time.

cleaning bot



for the child



building & making



name the parts

on the picture, circle each part and write its name next to it.



label these four parts

the wheels' motor

the brush's motor

the battery

the brush

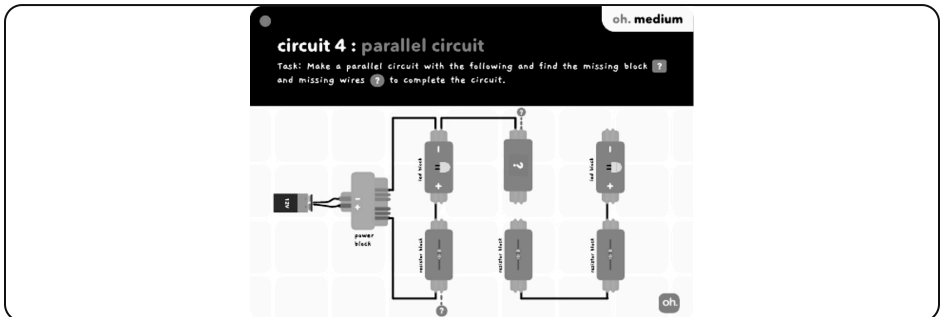


bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when you switched it on?

- ☐ the wheels AND the brush both worked together
- ☐ only one worked, never both
- ☐ nothing moved

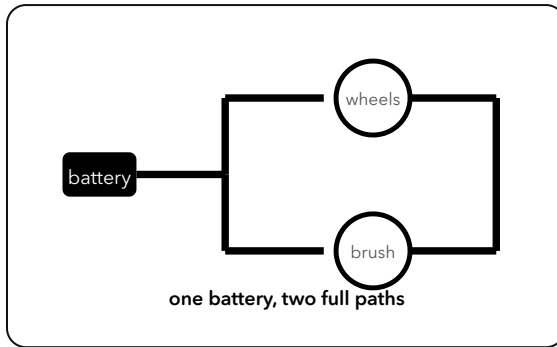


draw what you saw



observing & understanding

the big idea – why does it work?



the power from **one battery** splits and runs **both** motors at once – each along its own path – so the bot can **drive and sweep** at the same time.



can you finish it?

one battery in your bot can run...



two jobs at the same time



only ever one job



problem solving



how could you make it even better?

circle the idea that works best.

1 • reach dust in the corners

which idea would help?

- ☐ add a small brush that sticks out to the side
- ☐ make the bot wider
- ☐ remove the brush

2 • keep going for longer

what would help most?

- ☐ a fresh battery with more power
- ☐ a smaller battery
- ☐ no battery

3 • clean and drive without stopping

how does your bot already do this?

- ☐ one battery powers both jobs at once
- ☐ it does one job then the other
- ☐ it can only drive



your idea: think of one more way to make it better. draw it and label it.

draw your idea here



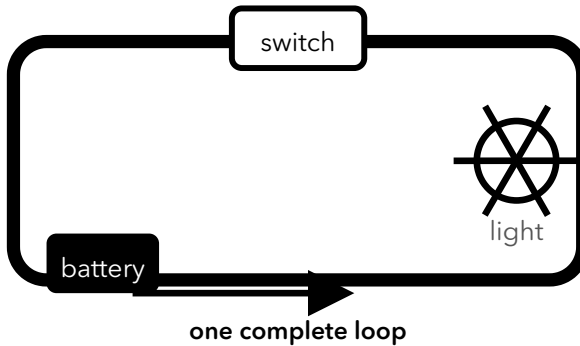
problem solving



engineer challenge – design your own

you want **one** battery to run the wheels **AND** the brush together.

here's an example – a switch turns a light on and off:



🔧 tick the blocks you would need

power block

motor block (wheels)

motor block (brush)

wires

switch block



now draw your own circuit – join the blocks in a **complete loop** so it works.

🔧 draw your circuit here



presenting

can you be the teacher?

on.



show a partner your cleaning bot. explain:

1. what does it do?
2. which part gives it power?
3. which part makes it work?



challenge: show them how to drive and sweep at once.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it sweeps the floor while it drives around.)

family member: _____

signature: _____



meet the sensor-controlled crane



your sensor-controlled crane – it lifts the load when the sensor is triggered.

At the port, huge cranes lift heavy boxes off the ships. Rehan notices something clever – the crane seems to KNOW when a box is under it, and only then does it lift. how can a machine sense and then act all by itself? let's build one.

words to know

sensor

input

output



did you know?

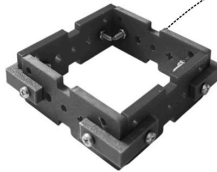
a machine part that **senses** the world (like light or an object) is called a **sensor**.



building & making

build your sensor-controlled crane – follow the steps.

step 1 : assemble the base

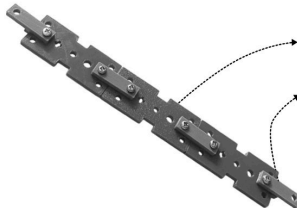


Let us start by building the base of the Crane.

Join 2 x F-F Beam and 2 x M-M Beam together to form a square.

Use 4 x U Clamps to hold them together.

step 2: assemble the supports



Join 3 x M-F Beam together using 1 clamps as shown.

Attach two extra 1 clamps to the ends of the pillar as shown

 Repeat this step to make another set.

step 3: attach the supports to the base



Attach the 2 supports to the base as shown.



tick when you've built the frame

 next: wire the sensor to the motor, so it lifts the load when the sensor is triggered.



building & making



name the parts



on the picture, circle each part and write its name next to it.

oh.



label these four parts

the sensor

the motor

the hook

the battery

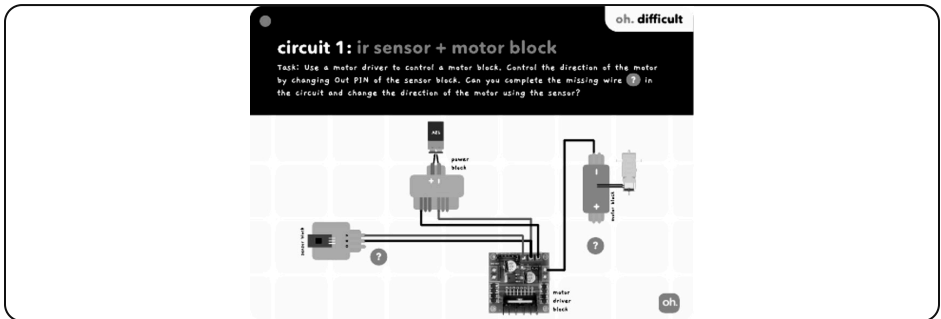


bonus: draw an arrow to show how it moves.



observing & understanding

build this circuit and watch closely.



what to look for

1. find the switch or motor and turn it on
2. watch what your machine does
3. now turn it off – what changes?

what happened when something came near the sensor?

- ☐ the sensor noticed it and the motor lifted the load
- ☐ nothing happened at all
- ☐ the battery ran out

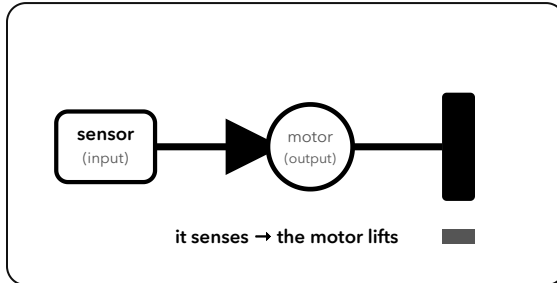


draw what you saw



observing & understanding

the big idea – why does it work?



the **sensor** is the **input** – it senses when something is there. the **motor** is the **output** – it acts. the input tells the output what to do.

 **can you finish it?**

the sensor's job is to...

- ☐ sense, then tell the motor to act
- ☐ do nothing at all



problem solving



how could you make it even better?

circle the idea that works best.

1 • lift a heavier box

which idea would help?

- ☐ use a pulley so it lifts with less effort
- ☐ use a thinner string
- ☐ take the motor off

2 • stop before it drops the load

what would help?

- ☐ a sensor that tells the motor when to stop
- ☐ no sensor
- ☐ close your eyes

3 • know when something is in the way

what does the sensor do?

- ☐ it senses the object and signals the motor
- ☐ it makes noise for fun
- ☐ nothing useful



your idea: think of one more way to make it better. draw it and label it.

draw your idea here



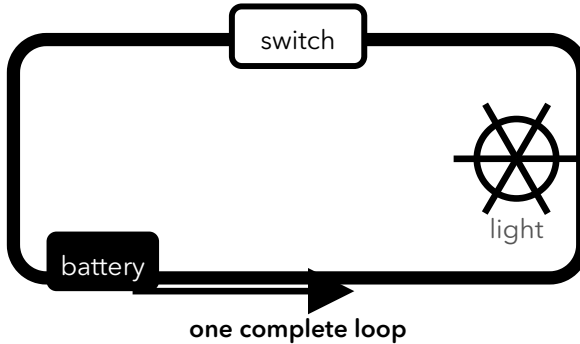
problem solving



engineer challenge – design your own

you want the **sensor** to make the **motor lift** when it senses a load.

here's an example – a switch turns a light on and off:



🔧 tick the blocks you would need

power block

sensor block

motor block

pulley & hook

wires



now draw your own circuit – join the blocks in a **complete loop** so it works.

🔧 draw your circuit here



presenting

can you be the teacher?



show a partner your sensor-controlled crane. explain:

1. what does it do?
2. which part gives it power?
3. which part makes it work?

★ **challenge:** show them how to trigger the sensor to lift the load.



at-home show & tell

take it home and show a family member. explain: **what problem does it solve?**
how does it work? (it senses a load and lifts it, all by itself.)

family member: _____

signature: _____